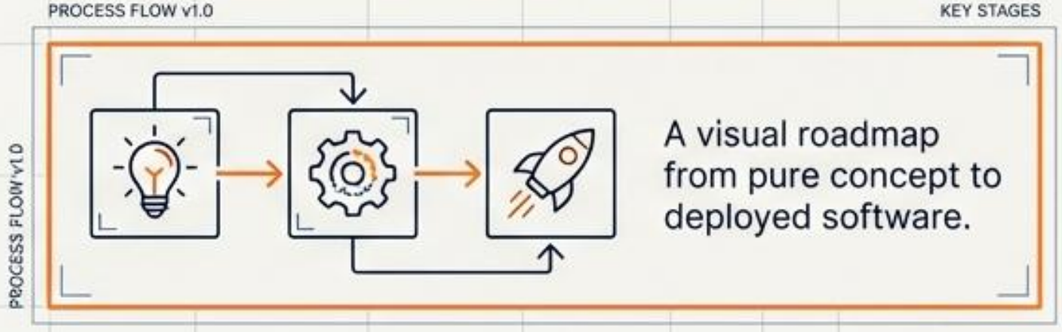
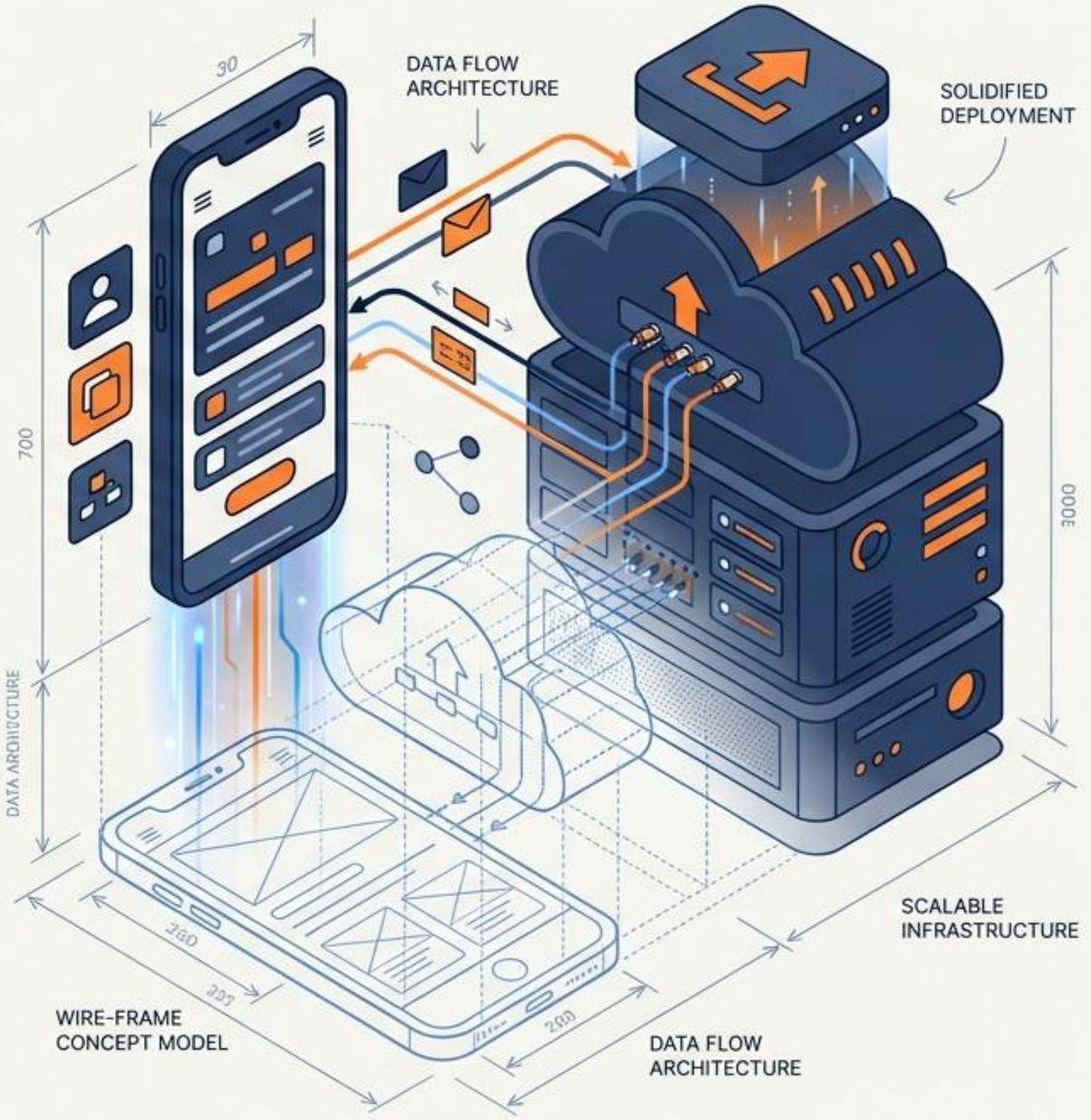
Application Development Foundation

Your blueprint for transforming ideas into reality.

Semester 2 2025/2026

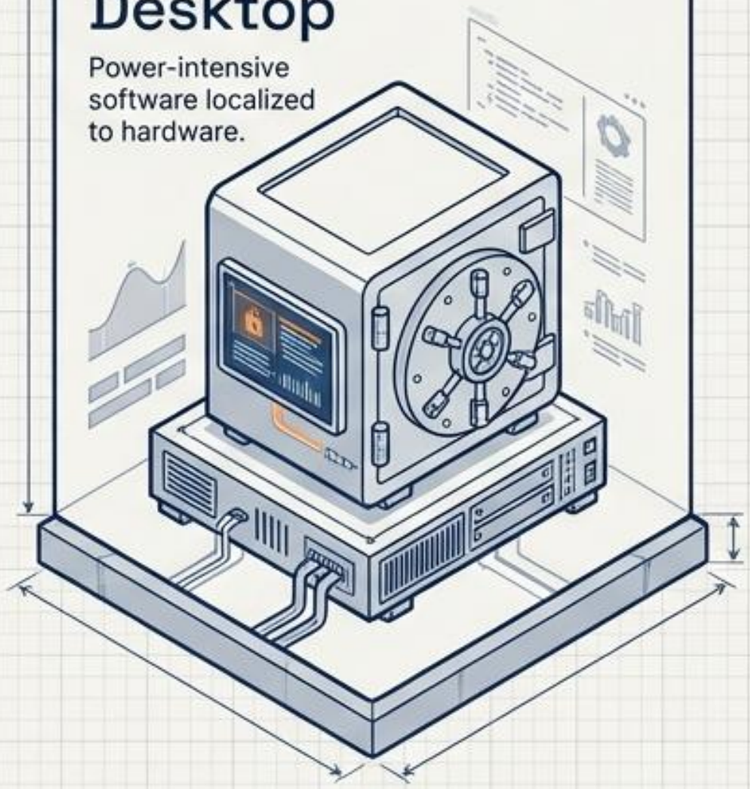




The Application Ecosystem

Desktop

Power-intensive software localized to hardware.



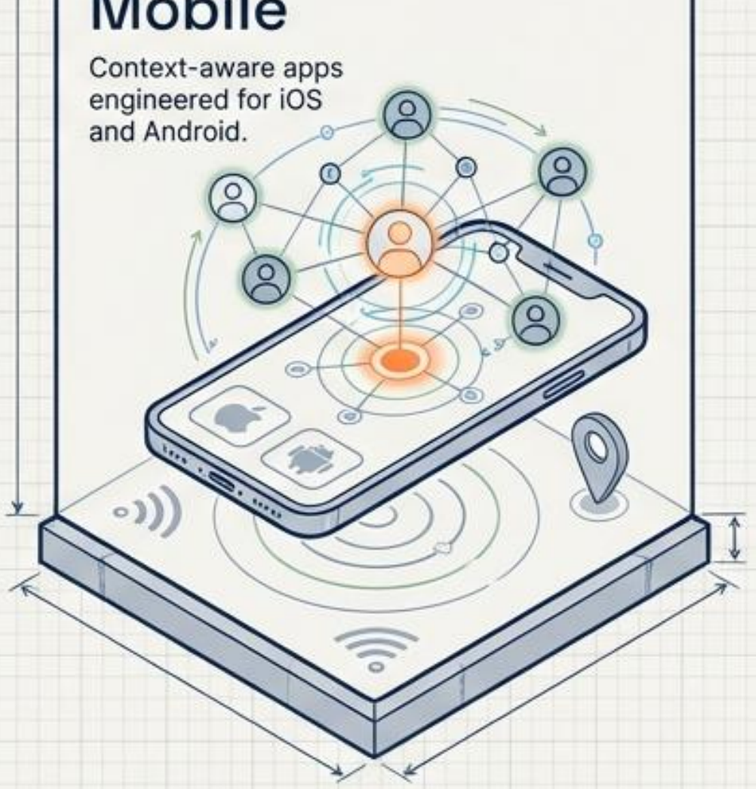
Web

Universally accessible, browser-based platforms.



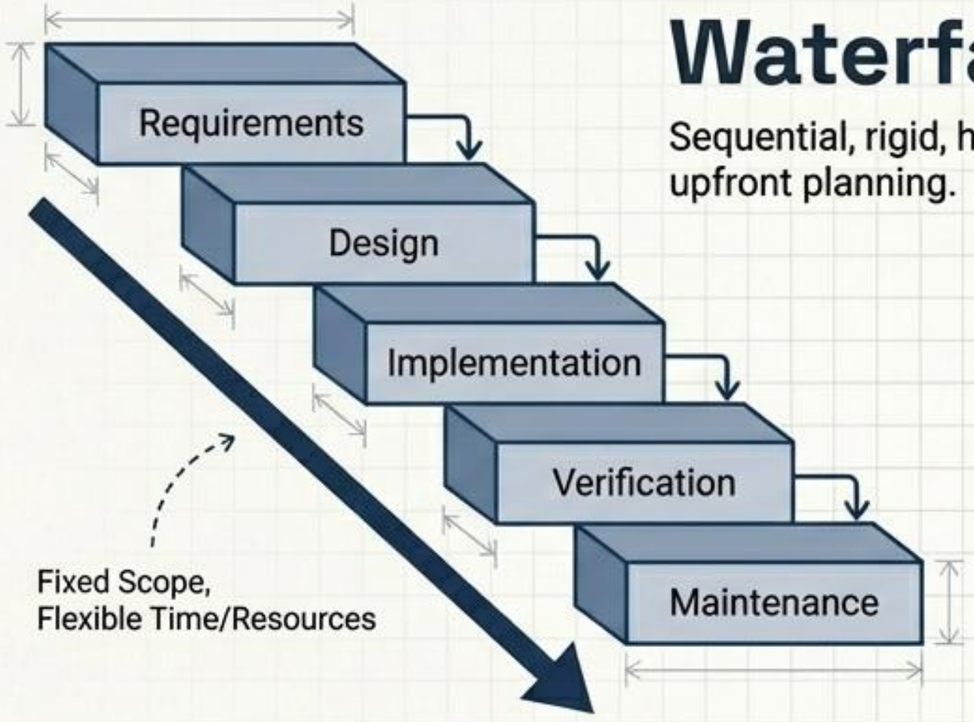
Mobile

Context-aware apps engineered for iOS and Android.



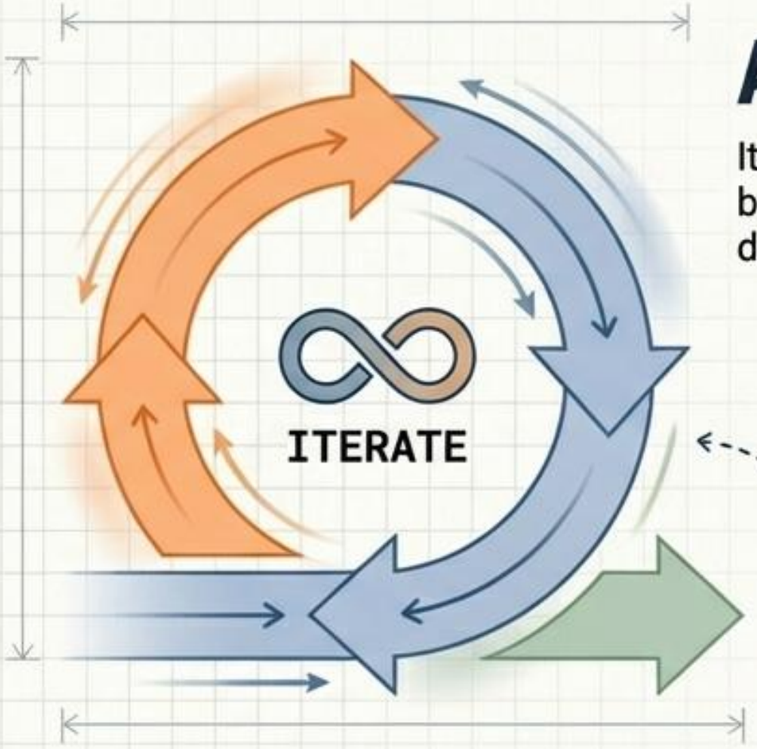
Why it matters — Applications are the connective tissue of modern business, fundamentally reshaping banking, commerce, and human interaction.

The Collaborative Engine: Development Methodologies



Waterfall

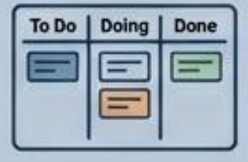
Sequential, rigid, heavy upfront planning.



Agile

Iterative, adaptable, built on continuous delivery.

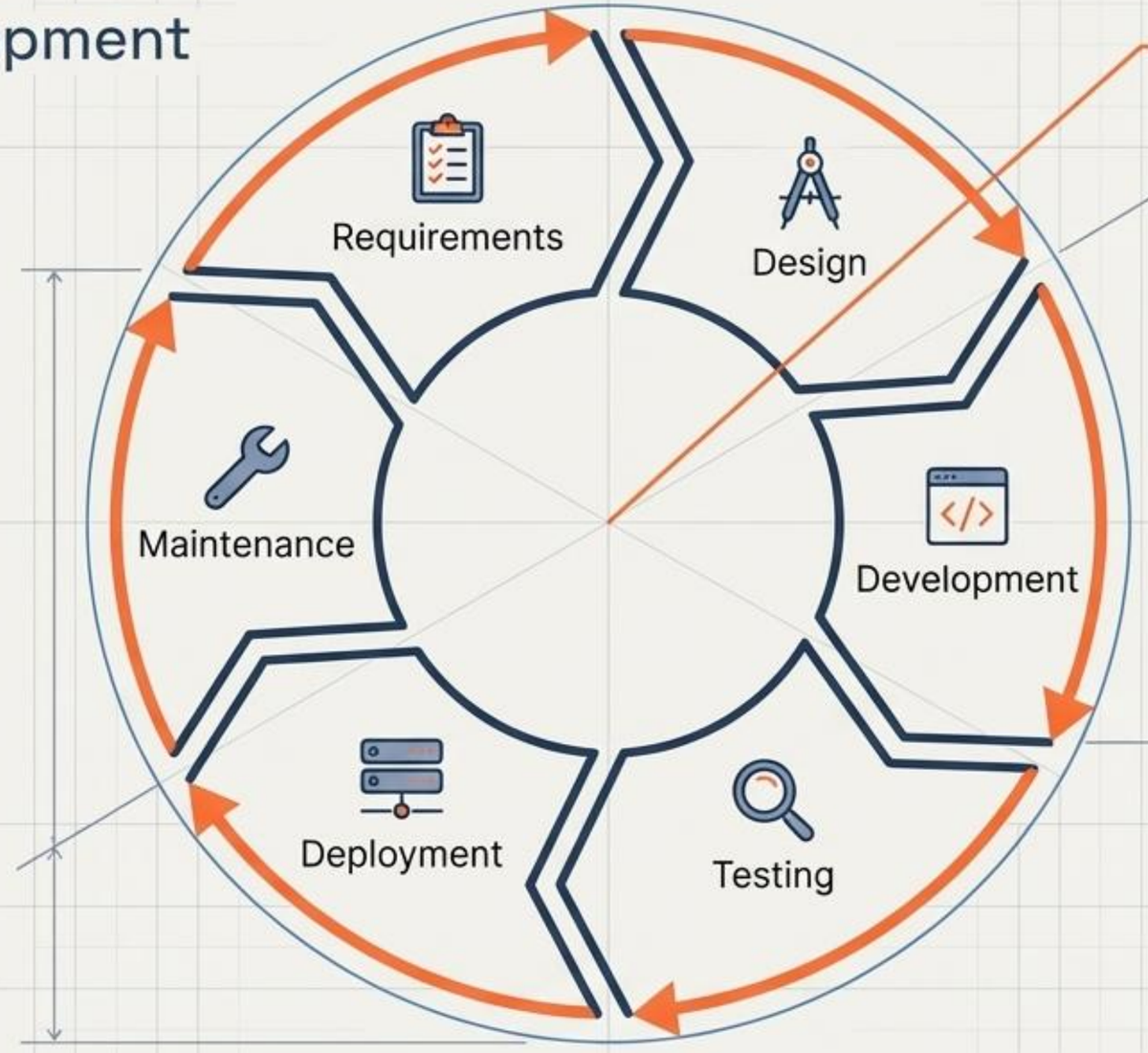






The Scrum Ecosystem: Defined roles, specific ceremonies, managed entirely within collaboration hubs like Jira and Trello.

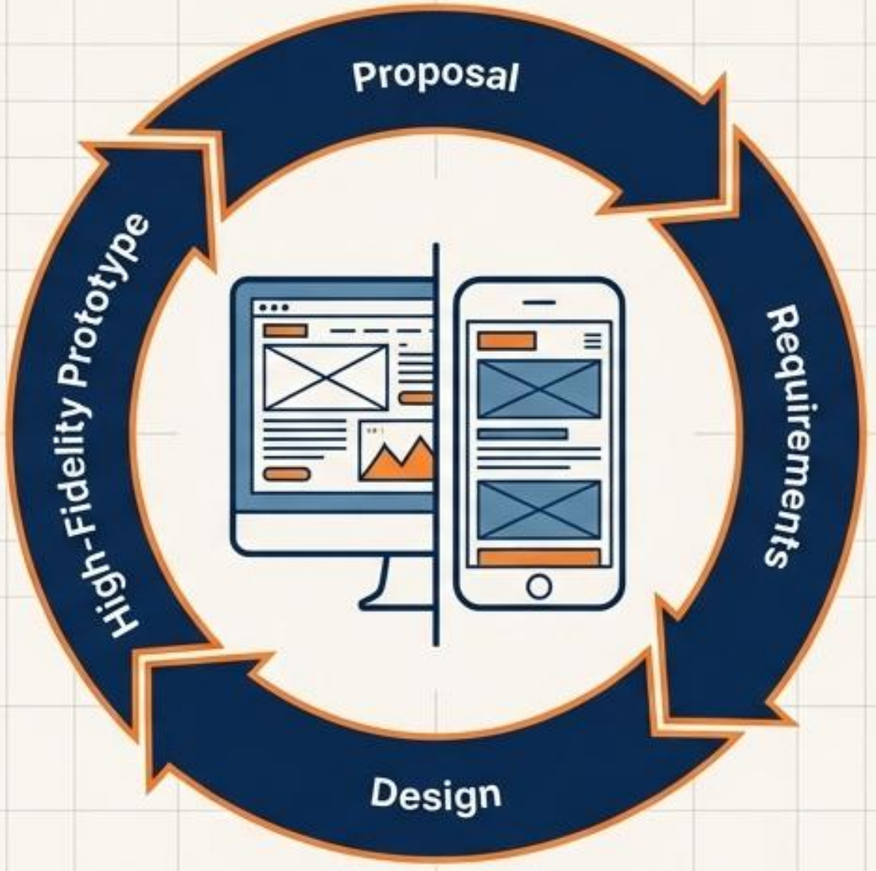
The Master Plan: Software Development Life Cycle



The Master Plan

Iteration & Feedback. Building software is never a straight line; continuous feedback loops dictate success.

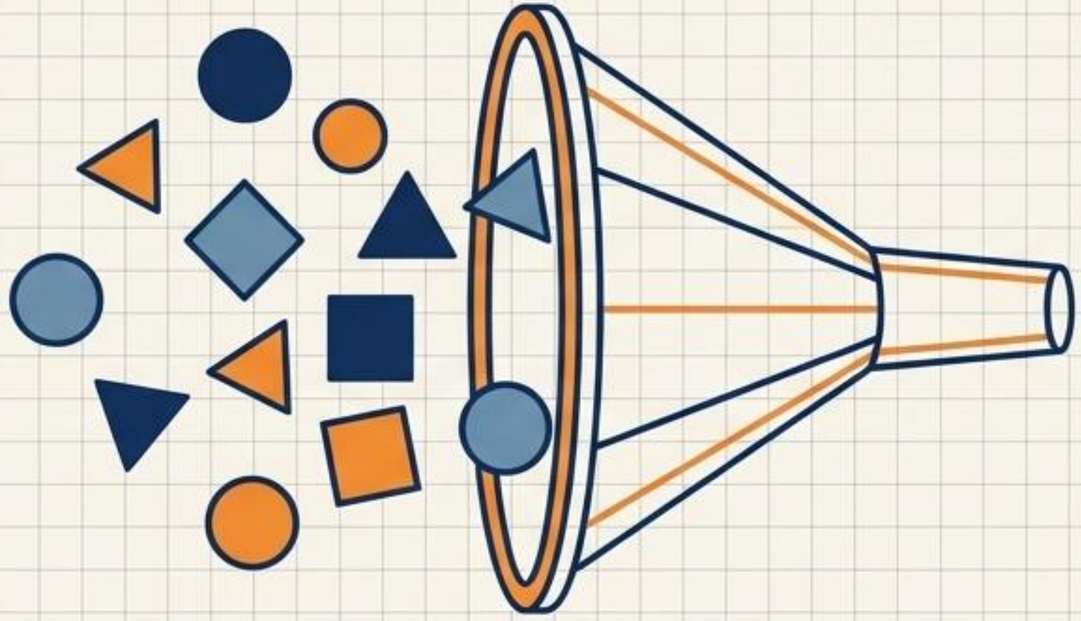
Executing the Full-Cycle Methodology



Standardizing detail stage development ensures absolute alignment across ecosystems. From initial proposal to the final high-fidelity prototype, Web and Mobile applications share a singular, proven path to realization.

Stage 1: Proposal

01



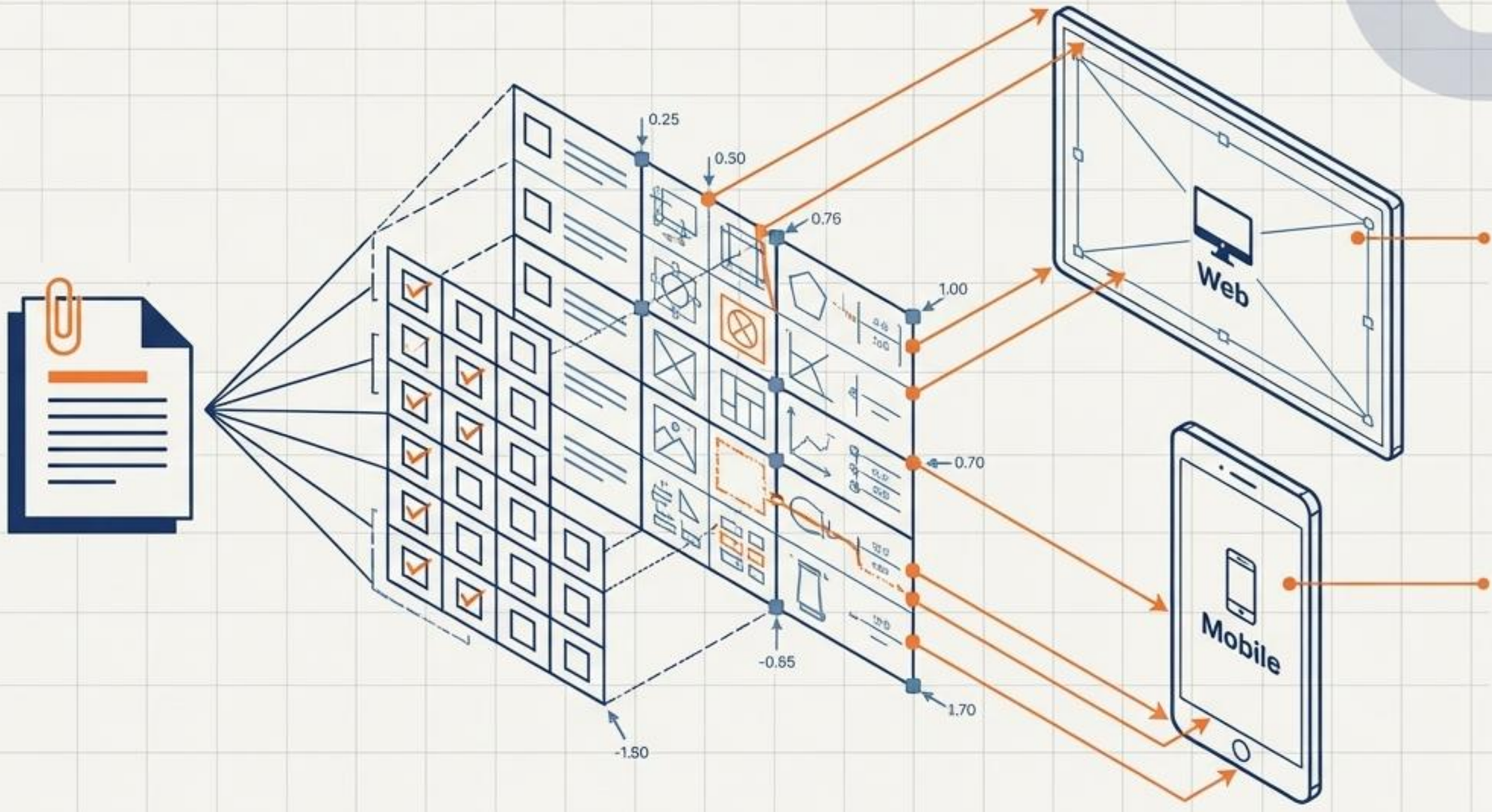
Web applications detail stage development start from proposal.

Mobile applications detail stage development start from proposal.

The Proposal acts as the universal catalyst, defining the initial scope and intent before diverging into platform-specific technical execution.

Stage 2: Requirements

02

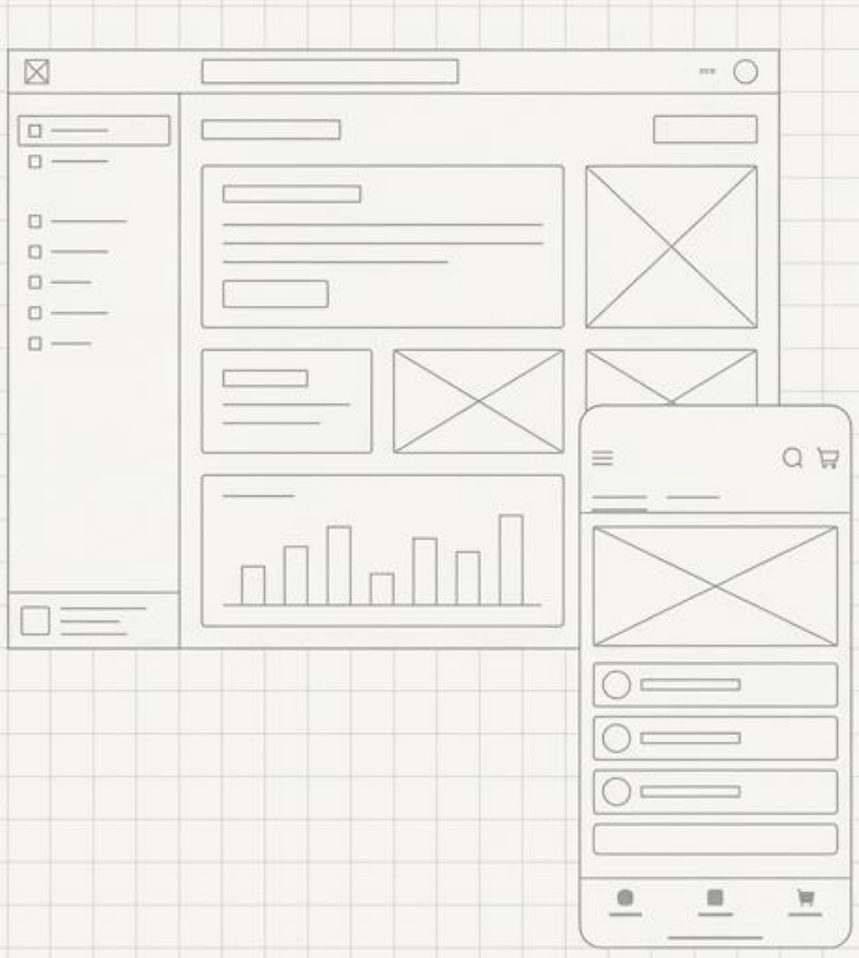


The progression dictates that Web applications move sequentially from proposal into rigid requirements.

Mobile applications follow the exact same path, moving directly from proposal to formalized requirements.

Stage 3: Design

03





JetBrains Mono
Design execution for
Web applications.

JetBrains Mono
Design execution for
Mobile applications.

Following requirements, the design stage dictates the visual and interactive language for both full-scale web interfaces and compact mobile environments.

The Human Element: UI & UX Design



Industry-standard design happens in collaborative platforms like Figma and Visual Paradigm.

Stage 4: High-Fidelity Prototype

04

The definitive culmination of detail stage development.



Both Web applications and Mobile applications finalize this sequence by delivering a high-fidelity prototype, effectively concluding the design progression.

High-Fidelity Prototyping and Technical Integration

A standardized overview of the design-to-development handoff.



Discover Our Solutions

Streamline your workflow with integrated tools designed for efficiency and growth.

Explore Platform

```
<div class="card-container">
  <div class="image-placeholder"></div>
  <div class="content">
    <h2>Discover Our Solutions</h2>
    <p>Streamline your workflow...</p>
  </div>
  <button class="primary-button">Explore Platform</button>
</div>

.card-container {
  background: #FFFFFF;
  border-radius: 12px;
  box-shadow: 0 4px 8px rgba(0,0,0,0.1);
}
.primary-button {
  background-color: #607D8B;
  color: #FFFFFF;
} ...
```

Evaluating the final design artifact in the later stages of design.

Assessing the high-fidelity prototype for completeness and realism.

Detailed

Complex data chart and typography hierarchy with clear visual priority.



Interactive

Prominent action button and intuitive navigation for user engagement.

Visually Polished

High-resolution hero image and refined layout for professional appeal.

The hi-fi prototype must closely resemble the final, coded application.

Prototype testing secures alignment and mitigates risk before development.

Must be completed before development initiates



Validate user experience



Uncover usability issues



Secure stakeholder approval

Transitioning from an approved prototype to version-controlled code

